

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

Lab 4 - SMTP Email Traffic Forensics

Lab Title:	SMTP Email Traffic Forensics
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Student ID:	2025/FWSD/11339
Course Code:	SBT-DF203
Course Name:	Basic Networking Skills for Digital Forensics

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1. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux or Ubuntu VM	Analyze the capture.
Packet tools	Wireshark and tshark	Filter and reconstruct SMTP streams.
Decoder	Python 3 base64 module	Decode Base64 strings offline.
Hashing	sha256sum	Preserve evidence integrity.
Evidence	smtp.pcap	Historical training traffic.
Text editor	Any local editor	Document recovered message safely.

2. Lab Setup

```
(kali@kali)-[~/Desktop]
└─$ mkdir -p ~/SBT-DF203-Lab4/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab4
pwd
/home/kali/SBT-DF203-Lab4
find . -maxdepth 1 -type d -print
./exported
./evidence
./working
./screenshots
./scripts
./reports
```

```
(kali@kali)-[~/SBT-DF203-Lab4]
└─$ wget -O evidence/smtp.pcap \
  'https://wiki.wireshark.org/uploads/_moin_import_/attachments/SampleCaptures/smtp.pcap'
--2026-09-14 17:50:33-- https://wiki.wireshark.org/uploads/_moin_import_/attachments/SampleCaptures/smtp.pcap
Resolving wiki.wireshark.org (wiki.wireshark.org) ... 104.26.11.240, 172.67.75.39, 104.26.10.240, ...
Connecting to wiki.wireshark.org (wiki.wireshark.org)|104.26.11.240|:443 ... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://gitlab.com/wireshark/wireshark/-/wikis/uploads/_moin_import_/attachments/SampleCaptures/smtp.pcap [following]
--2026-09-14 17:50:41-- https://gitlab.com/wireshark/wireshark/-/wikis/uploads/_moin_import_/attachments/SampleCaptures/smtp.pcap
Resolving gitlab.com (gitlab.com) ... 172.65.251.78, 2606:4700:90:0:f22e:fbec:5bed:a9b9
Connecting to gitlab.com (gitlab.com)|172.65.251.78|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 27850 (27K) [application/octet-stream]
Saving to: 'evidence/smtp.pcap'

evidence/smtp.pcap      100%[=====>] 27.20K  --.-KB/s  in 0.05s

2026-09-14 17:50:42 (577 KB/s) - 'evidence/smtp.pcap' saved [27850/27850]
```

```
capinfos evidence/smtp.pcap | tee reports/smtp_capinfos.txt
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab evidence/smtp.pcap
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab working/smtp_working.pcap
File name: evidence/smtp.pcap
File type: Wireshark/tcpdump/ ... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 65535 bytes
Number of packets: 60
File size: 27 kB
Data size: 26 kB
Capture duration: 9.198384 seconds
Earliest packet time: 2009-10-05 02:06:07.492060
Latest packet time: 2009-10-05 02:06:16.690444
Data byte rate: 2,920 bytes/s
Data bit rate: 23 kbps
Average packet size: 447.77 bytes
Average packet rate: 6 packets/s
SHA256: 17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
SHA1: 3def1a1fed849e7f23b66e6925dfff05845a400b
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
Encapsulation = Ethernet (1 - ether)
Capture length = 65535
Time precision = microseconds (6)
Time ticks per second = 1000000
Number of stat entries = 0
Number of packets = 60
```

3. Part A - Inventory the Capture and Locate SMTP Streams

```
(kali@kali) [~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -q -z conv,tcp | tee reports/tcp_conversations.txt

TCP Conversations
Filter:<No Filter>

| Frames | Bytes | | Frames | Bytes | | Total | | Relative | | Duration | |
| Frames | Bytes | | Frames | Bytes | | Frames | Bytes | | Start | | |
10.10.1.4:1470 ↔ 74.53.140.153:25 25 1,980 bytes 28 22 kB 53 24 kB 0.036986000 7.5777
```

tshark -r working/smtp_working.pcap -q -z conv,tcp | tee reports/tcp_conversations.txt

Output

-z conv,tcp asks TShark's built-in statistics engine to summarize every TCP conversation in the file: source/destination IP:port pairs, packet counts, and byte totals.
-q suppresses the normal per-packet listing so only the summary prints. This is how you find which conversation is the SMTP session

```
20 2009-10-05T02:06:10.320203000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA
21 2009-10-05T02:06:10.661679000-0400 74.53.140.153 25 10.10.1.4 1470 S: 354 Enter message, ending with "." on a line
self
22 2009-10-05T02:06:10.692743000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1460 bytes
23 2009-10-05T02:06:10.692786000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1460 bytes
24 2009-10-05T02:06:10.692804000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1460 bytes
25 2009-10-05T02:06:10.692823000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Window Full] C: DATA fragment, 1460 bytes
26 2009-10-05T02:06:10.695115000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragment
n needed)
28 2009-10-05T02:06:10.695623000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragment
n needed)
29 2009-10-05T02:06:10.696248000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragment
n needed)
30 2009-10-05T02:06:10.696634000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragment
n needed)
32 2009-10-05T02:06:11.104972000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
33 2009-10-05T02:06:11.104998000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
35 2009-10-05T02:06:11.469759000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
36 2009-10-05T02:06:11.469814000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
38 2009-10-05T02:06:11.496181000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
39 2009-10-05T02:06:11.496199000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
41 2009-10-05T02:06:11.834628000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
42 2009-10-05T02:06:11.834655000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
44 2009-10-05T02:06:11.858316000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
45 2009-10-05T02:06:11.858334000-0400 10.10.1.4 1470 74.53.140.153 25 from: "Gurpartap Singh" <gurpartap@patriots.in>,
ect: SMTP, (text/plain) (text/html) (text/plain) (text/plain) | .
52 2009-10-05T02:06:12.248789000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250 OK id=1Mugho-0003Dg-Un
54 2009-10-05T02:06:14.763825000-0400 10.10.1.4 1470 74.53.140.153 25 C: QUIT
56 2009-10-05T02:06:15.105467000-0400 74.53.140.153 25 10.10.1.4 1470 S: 221 xc90.websitewelcome.com closing connection
```

tshark -r working/smtp_working.pcap -Y "smtp" -T fields \
-e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e
_ws.col.Info \
| tee reports/smtp_packet_inventory.tsv

Output

The -Y "smtp" display filter isolates only packets TShark has already identified as SMTP protocol traffic (it recognizes the protocol automatically from port 25 and command syntax). The extracted fields gives every SMTP-tagged frame's number, timestamp, endpoints, and TShark's one-line interpretation (_ws.col.Info). This single file gives the server port, client ephemeral port, first/last packet time, and total SMTP frame count needed below.

Item	Your Finding
SMTP server IP : port	74.53.140.153 : 25
Client IP : ephemeral port	10.10.1.4 : 1470
First SMTP packet time	2009-10-05T02:06:12.248789000-0400
Last SMTP packet time	2009-10-05T02:06:15.103667000-0400

Item	Your Finding
Total number of SMTP frames	5

4. Part B - Identify Commands and Response Codes

Seq	Direction	Command/Code	Meaning	Timestamp
1	Server → client	220	Service ready	2009-10-05T02:06:08.219663000-0400
2	Client → server	EHLO/HELO	Client greeting	2009-10-05T02:06:08.224809000-0400
3	Client → server	AUTH	Authentication method	2009-10-05T02:06:08.568729000-0400
4	Client → server	MAIL FROM	Envelope sender	2009-10-05T02:06:09.614414000-0400
5	Client → server	RCPT TO	Envelope recipient	2009-10-05T02:06:09.957250000-0400
6	Client → server	DATA	Message content begins	2009-10-05T02:06:10.320203000-0400
7	Client → server	QUIT	Session termination	2009-10-05T02:06:14.763825000-0400

```
(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y "smtp.req || smtp.rsp" -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst \
-e smtp.req.command -e smtp.req.parameter \
-e smtp.response.code \
| tee reports/smtp_commands_responses.tsv
6      2009-10-05T02:06:08.219663000-0400      74.53.140.153      10.10.1.4
7      2009-10-05T02:06:08.224809000-0400      10.10.1.4          74.53.140.153      EHLO      GP      220
9      2009-10-05T02:06:08.566183000-0400      74.53.140.153      10.10.1.4
10     2009-10-05T02:06:08.568729000-0400      10.10.1.4          74.53.140.153      AUTH      LOGIN   334
11     2009-10-05T02:06:08.911081000-0400      74.53.140.153      10.10.1.4
12     2009-10-05T02:06:08.911655000-0400      10.10.1.4          74.53.140.153
13     2009-10-05T02:06:09.253544000-0400      74.53.140.153      10.10.1.4          334
14     2009-10-05T02:06:09.254118000-0400      10.10.1.4          74.53.140.153
15     2009-10-05T02:06:09.613798000-0400      74.53.140.153      10.10.1.4          235
16     2009-10-05T02:06:09.614414000-0400      10.10.1.4          74.53.140.153      MAIL      FROM: <gurpartap@patriots.in>
17     2009-10-05T02:06:09.956765000-0400      74.53.140.153      10.10.1.4          250
18     2009-10-05T02:06:09.957250000-0400      10.10.1.4          74.53.140.153      RCPT      TO: <raj_deol2002in@yahoo.co.in>
19     2009-10-05T02:06:10.319708000-0400      74.53.140.153      10.10.1.4          250
20     2009-10-05T02:06:10.320203000-0400      10.10.1.4          74.53.140.153      DATA
21     2009-10-05T02:06:10.661679000-0400      74.53.140.153      10.10.1.4          354
52     2009-10-05T02:06:12.248789000-0400      74.53.140.153      10.10.1.4          250
54     2009-10-05T02:06:14.763825000-0400      10.10.1.4          74.53.140.153      QUIT
56     2009-10-05T02:06:15.105467000-0400      74.53.140.153      10.10.1.4          221
```

tshark -r working/smtp_working.pcap -Y "smtp.req || smtp.rsp" -T fields \

```
-e frame.number -e frame.time -e ip.src -e ip.dst \
-e smtp.req.command -e smtp.req.parameter \
-e smtp.response.code \
| tee reports/smtp_commands_responses.tsv
```

Output

smtp.req matches client-issued commands; smtp.rsp matches server responses. Extracting smtp.req.command/parameter alongside smtp.response.code/parameter and gives a clean, ordered table of exactly what the client asked for and how the server replied, this is the raw material for the table above and session-timeline section.

5. Part C - Decode Base64 Authentication Evidence

frames **12 and 14** are client → server packets following the two 334 responses. They are likely the Base64-encoded authentication values which i nvestigate earlier to know how to handle the **AUTH LOGIN/Base64** section

```
tshark -r working/smtp_working.pcap -Y 'frame.number==12 || frame.number==14' -V
```

After my investigation with tshark two Base64 values found



```
(kali@kali)-[~/SBT-DF203-Lab4]
$ python3 - <<'PYLAB'

import base64

samples = {
    'username': 'Z3VycGFydGFwQHBhdHJpb3RzLmlu',
    'password': 'cHVuamFiQDEyMw=='
}

for label, value in samples.items():
    try:
        decoded = base64.b64decode(value).decode('utf-8', errors='replace')
        print(f'{label}: {decoded}')
    except Exception as exc:
        print(f'{label}: decode failed: {exc}')

PYLAB
username: gurpartap@patriots.in
password: punjab@123
```

Output

This is a small offline Python heredoc script, it never touches the network. base64.b64decode() reverses the Base64 encoding back to the original bytes, and .decode('utf-8', errors='replace') turns those bytes into readable text (substituting a placeholder character for anything that isn't valid UTF-8, so the script doesn't crash on odd input). once i pasted in the real encoded values were found in the capture.

6. Part D - Reconstruct the Email Message

1. In Wireshark (GUI), right-click any SMTP packet in the session and choose Follow → TCP Stream. This reassembles the full client/server conversation in order, in plain text.
2. Identify SMTP envelope fields (MAIL FROM / RCPT TO) separately from the RFC 5322 message headers found inside the DATA section — these are not always the same address.
3. Record the Date, From, To, Subject, Message-ID, MIME-Version, Content-Type and User-Agent/X-Mailer headers where present.
4. Determine whether the body is plain text, HTML, multipart, or contains an attachment (look for Content-Type: multipart/... and Content-Disposition: attachment).
5. Save a redacted text reconstruction to reports/reconstructed_email_redacted.txt — mask any personal data not essential to the training exercise.

To exact stream number

```
(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y smtp -T fields \
-e tcp.stream -e frame.number -e ip.src -e ip.dst -e tcp.srcport -e tcp.dstport \
| head -30
0      6      74.53.140.153    10.10.1.4      25      1470
0      7      10.10.1.4          74.53.140.153  1470     25
0      9      74.53.140.153     10.10.1.4      25      1470
0     10      10.10.1.4          74.53.140.153  1470     25
0     11      74.53.140.153     10.10.1.4      25      1470
0     12      10.10.1.4          74.53.140.153  1470     25
0     13      74.53.140.153     10.10.1.4      25      1470
0     14      10.10.1.4          74.53.140.153  1470     25
0     15      74.53.140.153     10.10.1.4      25      1470
0     16      10.10.1.4          74.53.140.153  1470     25
0     17      74.53.140.153     10.10.1.4      25      1470
0     18      10.10.1.4          74.53.140.153  1470     25
0     19      74.53.140.153     10.10.1.4      25      1470
0     20      10.10.1.4          74.53.140.153  1470     25
0     21      74.53.140.153     10.10.1.4      25      1470
0     22      10.10.1.4          74.53.140.153  1470     25
0     23      10.10.1.4          74.53.140.153  1470     25
```

My output confirms the **TCP stream number is 0**.

Reconstruction of the TCP stream

```
(kali㉿kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,0 \
| tee reports/smtp_stream_0.txt
```

```
Follow: tcp,ascii
Filter: tcp.stream eq 0
Node 0: 10.10.1.4:1470
Node 1: 74.53.140.153:25
      181
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.

9
EHLO GP

      137
250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP

12
AUTH LOGIN

      18
334 VXNlcm5hbWU6

30
```

Display the reconstructed email manually including /DATA

```
56
354 Enter message, ending with "." on a line by itself

1460
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
        .boundary="=_NextPart_000_0004_01CA45B0.095693F0"
X-Mailer: Microsoft Office Outlook 12.0
Thread-Index: AcpFgem9BvjjZEDeR1Kh8i+hUyVo0A=
Content-Language: en-us
x-cr-hashedpuzzle: SeA= AAR2 ADaH BpiO C4G1 D1gW FNB1 FPKR Fn+W HFCP HnYJ J07s Kum6 KytW LfCI LjUt;1;cgBhAGoAXwBkAGUAbwBsADIAMAAwADIAaQBuAEAAeQBhAGgAbwBvAC4AYwBvAC4AaQBuAA=;Soshal_v1;7;{CAA37F59-1850-45C7-8540-AA27696B5398};ZwB1AHIAcABhAHIAcABhAHIAcABwAGEAdABYAGkAbwB0AHMALgBpAG4A;Mon, 05 Oct 2009 06:06:01 GMT;UwBNAFQAUA=
x-cr-puzzleid: {CAA37F59-1850-45C7-8540-AA27696B5398}

This is a multipart message in MIME format.

-----=_NextPart_000_0004_01CA45B0.095693F0
Content-Type: multipart/alternative;
        .boundary="=_NextPart_001_0005_01CA45B0.095693F0"
```

/Date:

```
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
.boundary="====_NextPart_000_0004_01CA45B0.095693F0"
X-Mailer: Microsoft Office Outlook 12.0
Thread-Index: AcpFgem9BvjJZEDeR1Kh8i+hUyVo0A=
Content-Language: en-us
x-cr-hasheDPuzzle: SeA= AAR2 ADaH BpIo C4G1 D1gW FNB1 FPKR Fn+W HFPC HnYJ J07s Kum6 KytW LfCI LjUt;1;cgBhAGoAXwBkAGUAbwBsADIAMAAwADIAaQBwAUEA
AeQbHAGgAbwBVAC4AYwBVAC4AaQBuAA=;Sosha1_v1;7;[CAA37F59-1850-45C7-8540-AA27696B5398];ZwB1AHIAcABhAHIAcABhAHAAQABwAGEAdABYAgKABwB0AHMALg8pAG4
A;Mon, 05 Oct 2009 06:06:01 GMT;UwBNAFQAUAA=
x-cr-puzzleid: {CAA37F59-1850-45C7-8540-AA27696B5398}

This is a multipart message in MIME format.

====_NextPart_000_0004_01CA45B0.095693F0
Content-Type: multipart/alternative;
.boundary="====_NextPart_001_0005_01CA45B0.095693F0"

====_NextPart_001_0005_01CA45B0.095693F0
Content-Type: text/plain;
.charset="us-ascii"
Content-Transfer-Encoding: 7bit

Hello

I send u smtp pcap file
```

Automatic headers extraction

```
(kali㉿kali)-[~/SBT-DF203-Lab4]
$ grep -Ei 'Date:|From:|To:|Subject:|Message-ID:|MIME-Version:|Content-Type:|User-Agent:|X-Mailer:' \
reports/smtp_stream_0.txt \
| tee reports/message_headers.txt
MAIL FROM: <gurpartap@patriots.in>
RCPT TO: <raj_deol2002in@yahoo.co.in>
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
X-Mailer: Microsoft Office Outlook 12.0
Content-Type: multipart/alternative;
Content-Type: text/plain;
Content-Type: text/html;
Content-Type: text/plain;
```

Header	Value Found
Date	Mon, 5 Oct 2009 11:36:07 +0530
From	"Gurpartap Singh" <gurpartap@patriots.in>
To	<raj_deol2002in@yahoo.co.in>
Subject	SMTP
Message-ID	<000301ca4581\$ef9e57f0\$cedb07d0\$@in>
MIME-Version	1.0
Content-Type	multipart/mixed
User-Agent / X-Mailer	Microsoft Office Outlook 12.0

7. Part E - Determine Client, Hosts and Network Metadata

```
(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y "smtp" -T fields \
  -e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e tcp.srcport \
  -e ip.dst -e tcp.dstport -e tcp.stream \
  | tee reports/smtp_network_metadata.tsv
6      1254722768.219663000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
7      1254722768.224809000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
9      1254722768.566183000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
10     1254722768.568729000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
11     1254722768.911081000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
12     1254722768.911655000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
13     1254722769.253544000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
14     1254722769.254118000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
15     1254722769.613798000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
16     1254722769.614414000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
17     1254722769.956765000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
18     1254722769.957250000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
19     1254722770.319708000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 74.53.140.153 25 10.10.1.4 1470 0
20     1254722770.320203000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
21     1254722770.661679000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
22     1254722770.692743000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
23     1254722770.692786000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
24     1254722770.692804000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
25     1254722770.692823000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
26     1254722770.695115000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
28     1254722770.695623000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
29     1254722770.696248000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
30     1254722770.696634000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
32     1254722771.104972000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
33     1254722771.104998000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
35     1254722771.469759000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
36     1254722771.469814000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
38     1254722771.494181000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
39     1254722771.494199000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
41     1254722771.834628000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
42     1254722771.834655000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
44     1254722771.858316000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
45     1254722771.858334000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
52     1254722772.248789000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
54     1254722774.763825000 00:1f:33:d9:81:60 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
56     1254722775.105467000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
```

```
tshark -r working/smtp_working.pcap -Y "smtp" -T fields \
  -e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e tcp.srcport \
  -e ip.dst -e tcp.dstport -e tcp.stream \
  | tee reports/smtp_network_metadata.tsv
```

Output

Extends the earlier inventory to include MAC addresses (eth.src/eth.dst) and the TCP stream index (tcp.stream) for every SMTP-tagged packet. This is the metadata an analyst needs to place the session on a specific network segment and timeline, independent of message content, useful even if the payload turns out to be encrypted.

```
(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
  -T fields -e frame.number -e tcp.stream -e smtp.req.parameter -e data-text-lines \
  | tee reports/client_indicators.tsv
22      0      Line-based text data (47 lines)
26      0      Line-based text data (12 lines)
```

```
tshark -r working/smtp_working.pcap -Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
  -T fields -e frame.number -e tcp.stream -e smtp.req.parameter -e data-text-lines \
  | tee reports/client_indicators.tsv
```

8. Part F - Encryption and Evidential Limitations

- Check whether STARTTLS is requested in the command sequence, and whether the payload immediately after it becomes unreadable/binary (indicating the session actually upgraded to TLS).
- If traffic is encrypted from that point on, document precisely what metadata still remains visible: endpoints (IP/MAC), ports, timing, and — if a TLS handshake is visible — certificate/session information and overall byte counts.
- Do not claim to know message content or credentials for any portion of the session that occurred after a genuine STARTTLS upgrade — the capture does not reveal it.
- Explicitly state that port number alone (e.g. port 25 or 587) does not prove whether a session is encrypted; the actual protocol negotiation and packet contents must be examined and cited.

```
(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y "smtp contains \"STARTTLS\"" -T fields \
  -e frame.number -e frame.time -e _ws.col.Info | tee reports/starttls_check.tsv
9      2009-10-05T02:06:08.566183000-0400      S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] | SIZE 52428800 | PIPELINING | AUT
H PLAIN LOGIN | STARTTLS | HELP
```

9. Required Findings Worksheet

The core results table

Question	Finding
When did the SMTP session start and end?	Start: 2009-10-05 02:06:08.219663 EDT (Frame 6, server 220 banner). End: 2009-10-05 02:06:15.105467 EDT (Frame 56, server 221 response).
Client IP/MAC and port	IP: 10.10.1.4 - Port: 1470. MAC: 00:e0:1c:3c:17:c2 (CradlePoint_3c:17:c2).
Server IP/MAC and port	IP: 74.53.140.153 - Port: 25. Server MAC was not shown in the packet details provided.
SMTP server banner	220 xc90.websiteswelcome.com - the server also advertised SIZE 52428800, PIPELINING, AUTH PLAIN LOGIN, STARTTLS, and HELP.
Client software	Microsoft Office Outlook 12.0 (X-Mailer header).
Authentication method	AUTH LOGIN. The authentication exchange included Base64-encoded username and password data.
Envelope sender and recipient	MAIL FROM: <gurpartap@patriots.in>, RCPT TO: <raj_deol2002in@yahoo.co.in>.
Message From/To/Subject	From: "Gurpartap Singh" <gurpartap@patriots.in> To: <raj_deol2002in@yahoo.co.in> Subject: SMTP.
Message body type	Multipart message. The capture shows multipart/mixed and multipart/alternative, with text/plain and text/html sections.
Attachment present?	
STARTTLS/TLS observed?	STARTTLS was advertised , but no STARTTLS negotiation was observed in the authentication exchange. AUTH LOGIN credentials were visible in the capture.
Key limitations	Analysis is based only on the captured TCP stream. Some message content may be incomplete or unavailable. The server MAC address was not established from the supplied packet details. Attachment presence requires checking the MIME sections for attachment headers.

10. Executive Summary

This report documents the forensic analysis of an SMTP packet capture (smtp_working.pcap, SHA-256: **17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab**) conducted as part of SBT-DF203 Lab 4.

The capture contains **36 SMTP frames** spanning a session between client **10.10.1.4:1470** and server **74.53.140.153:25**, beginning at **2009-10-05 02:06:08.219663 EDT** and ending at **2009-10-05 02:06:15.105467 EDT**. The session followed a standard SMTP command/response sequence including **220, EHLO, AUTH LOGIN, MAIL FROM, RCPT TO, DATA, and QUIT**.

Authentication was performed using **AUTH LOGIN**. The captured username and password values were Base64 encoded and were decoded offline for forensic analysis. The credential values are **masked in this report** to avoid unnecessary disclosure.

The reconstructed message was sent from **Gurpartap Singh** <gurpartap@patriots.in> to <raj_deol2002in@yahoo.co.in> with subject **“SMTP”**, formatted as **multipart content containing plain-text and HTML sections**. **An attachment has not yet been confirmed** from the evidence provided.

STARTTLS was advertised by the SMTP server, but STARTTLS negotiation was not observed in the captured session. The authentication exchange occurred in visible SMTP traffic, allowing the Base64-encoded authentication values to be recovered from the capture.

Key limitations of this analysis are: **the analysis is based on a single packet capture; server-side logs were not available; the server MAC address was not established from the supplied packet details; and attachment presence has not yet been confirmed from the MIME content.**